

Masonry

✓ Masonry refers to a construction material, which is formed by combining individual masonry units such as stone and brick, with a binding material i.e., mortar.

✓ Masonry is mainly used to build walls, which serve to enclose or divide a space and to support loads from other elements.

Manufacture of clay bricks

✓ Clays for manufacture of bricks differ widely in composition from place to place. Even in the same site, clay deposits from one part or depth may vary significantly from that from another part or depth.

✓ Clays are composed mainly of silica (grains of sand), alumina, lime, iron, manganese, sulfur and phosphates.

Types of clay:

✓ The types of clay used in the manufacture of bricks are as follows:

- Surface clay
- Shale
- Fire clay

Manufacture of clay bricks

✓ Bricks are manufactured by grinding or crushing the clay in mills, and mixing it with water to make it plastic. The mixing of clay with water is carried out in pug mills. The plastic clay is then moulded (in any of the three processes of forming i.e., stiff-mud process, soft-mud process and dry press process), textured, dried and finally fired. The firing temperature for manufacturing of bricks varies from 900 °C to 1200 °C with temperature as high as 1315 °C in burning kilns. After burning, the units are cooled.

Properties of clay bricks

Physical properties:

- Colour
- Texture
- Size
- Density
- Water absorption

Mechanical properties:

- Compressive strength
- Modulus of rupture
- Modulus of elasticity
- Tensile strength
- Thermal conductivity
- Fire resistance

Mortar for masonry

✓ Mortar is a plastic mixture of cementitious materials, sand and water, and it binds together individual masonry units. Mortar serves as a bedding material for the masonry units and allows the level placement of the units.

- **Lime mortar:** mixture of lime, sand, and water.

- **Cement lime mortar or cement mortar:** mixture of cement, lime, sand and water.

✓ The mortar can be prescribed using one of the following two alternative types of specifications:

- Proportion specifications
- Property specifications

Mortar cement: It is a hydraulic cement, primarily used in masonry construction. It consists of a mixture of Portland or blended hydraulic cement and plasticizing materials (such as limestone or hydrated or hydraulic lime) together with other materials introduced to enhance one or more properties such as setting time, workability, water retention and durability.

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TABLE 1 Property Specification Requirements^A

Mortar	Type	Average Compressive Strength at 28 days, min, psi (MPa)	Water Retention, min, %	Air Content, max, % ^B	Aggregate Ratio (Measured in Damp, Loose Conditions)
Cement-Lime ^C	M	2500 (17.2)	75	12	Not less than 2 1/4 and not more than 3 1/2 times the sum of the separate volumes of cementitious materials
	S	1800 (12.4)	75	12	
	N	750 (5.2)	75	14 ^D	
	O	350 (2.4)	75	14 ^D	
Mortar Cement	M	2500 (17.2)	75	18	
	S	1800 (12.4)	75	18	
	N	750 (5.2)	75	20 ^E	
	O	350 (2.4)	75	20 ^E	
Masonry Cement	M	2500 (17.2)	75	18	
	S	1800 (12.4)	75	18	
	N	750 (5.2)	75	20 ^E	
	O	350 (2.4)	75	20 ^E	

^ALaboratory prepared mortar only (see Note 5).

^BSee Note 6.

^CIncludes Specification C150/C150M, C595/C595M, and C1157/C1157M cements as described in 4.1.1.

^DWhen structural reinforcement is incorporated in cement-lime, the maximum air content shall be 12 %.

^EWhen structural reinforcement is incorporated in masonry cement mortar or mortar cement mortar, the maximum air content shall be 18 %.

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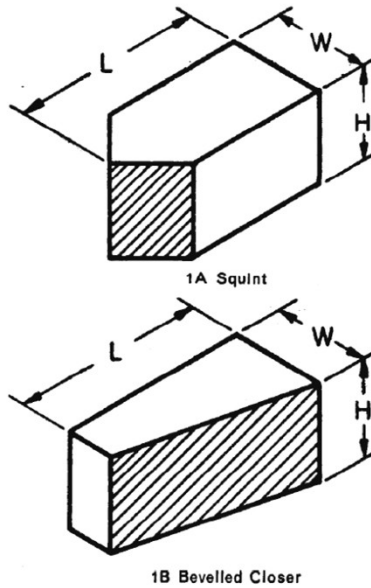
TABLE 2 Proportion Specification Requirements

NOTE 1—Two air-entraining materials shall not be combined in mortar.

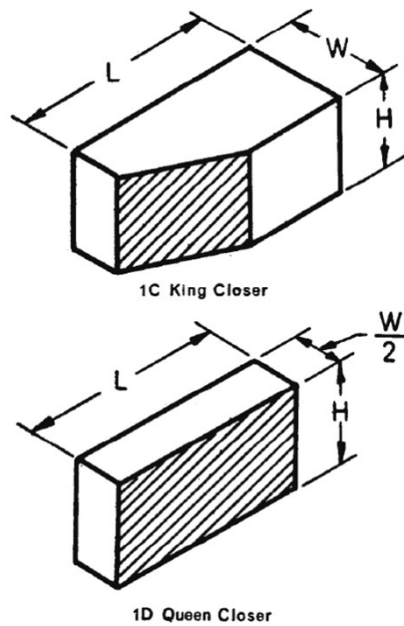
Mortar	Type	Cement ⁴	Proportions by Volume (Cementitious Materials)						Hydrated Lime or Lime Putty	Aggregate Ratio (Measured in Damp, Loose Con- ditions)
			Mortar Cement			Masonry Cement				
			M	S	N	M	S	N		
Cement-Lime	M	1	...	...	...	...	...	...	¼	Not less than 2¼ and not more than 3 times the sum of the separate vol- umes of cementi- tious materials
	S	1	...	...	...	...	...	...	over ¼ to ½	
	N	1	...	...	...	...	...	...	over ½ to 1¼	
	O	1	...	...	...	...	...	...	over 1¼ to 2½	
Mortar Cement	M	1	...	...	1	...	...	...	...	
	M	...	1	...	...	...	...	...	...	
	S	½	...	...	1	...	...	...	...	
	S	...	...	1	...	...	...	...	...	
	N	...	...	...	1	...	...	...	...	
	O	...	...	...	1	...	...	...	...	
Masonry Cement	M	1	...	...	...	...	...	1	...	
	M	...	...	...	...	1	...	...	...	
	S	½	...	...	...	...	...	1	...	
	S	...	...	...	...	...	1	...	...	
	N	...	...	...	...	...	...	1	...	
	O	...	...	...	...	...	...	1	...	

^AIncludes Specification C150/C150M, C595/C595M, and C1157/C1157M cements as described in 4.1.1.

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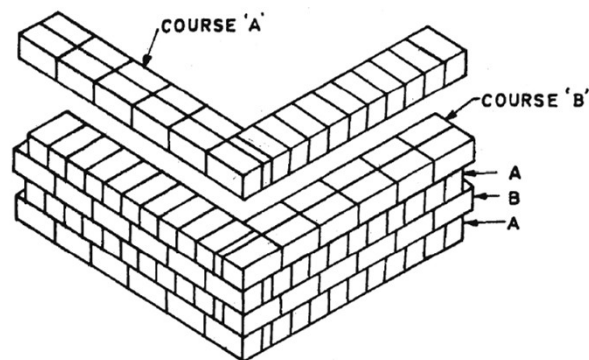


FIG. 2 ENGLISH BOND

- The bricks in the facing are laid in alternate courses of headers and stretchers. The header course is commenced with a quoin header followed by a queen closer and continued with successive headers. The stretcher course is formed of stretchers having a minimum lap of one-quarter their length over the header.
- This bond contributes substantially to the strength of brickwork and may be particularly suitable for walls carrying heavy loads.

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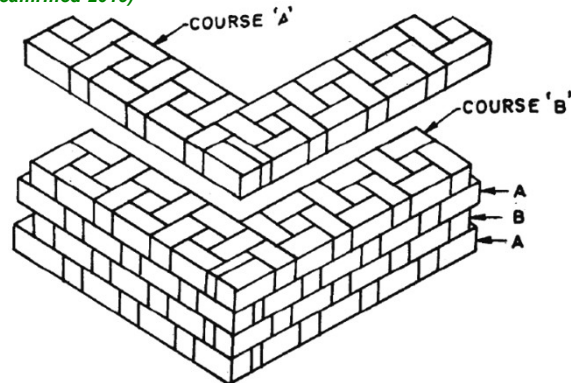


FIG. 3 DOUBLE FLEMISH BOND

- The courses both in the facing and in the backing are formed with alternating stretchers and headers. In order to obtain the lap (which is equal to one-quarter the length of the bricks), a queen closer is introduced next to the quoin header in alternate courses, the intervening ones commencing with a stretcher and every header will obtain a location that is central with respect to the stretcher above or below.
- The appearance on the face may be considered as more uniform than in English bond.

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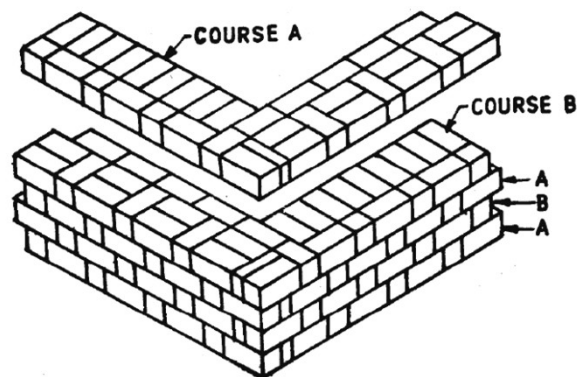


FIG. 4 SINGLE FLEMISH BOND

- This bond facilitates the facing of the wall to be in Flemish bond and the backing in English bond.
- This attempts to combine partially the better appearance of Flemish bond with better strength of English bond. However, the increase in strength over Double Flemish bond is doubtful.

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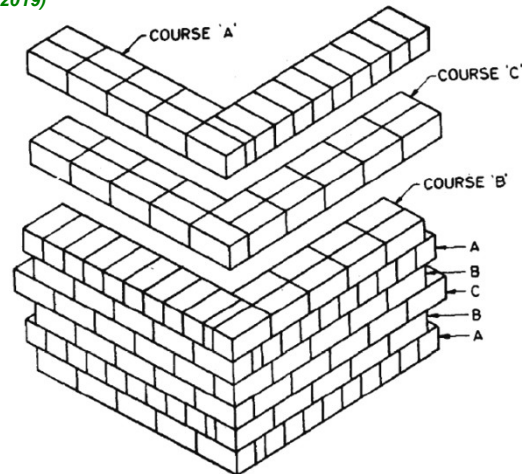


FIG. 5 ENGLISH GARDEN WALL BOND

- English garden wall bond consists of a header course with the necessary queen closer next to quoin header to three or sometimes even five stretcher courses running in series with overlap of half-brick between stretcher over stretcher.

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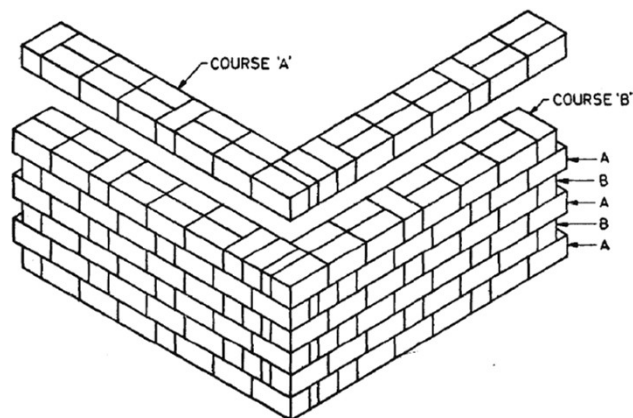


FIG. 6 FLEMISH GARDEN WALL BOND

- Flemish garden wall bond consists of alternate courses composed of one header to three, sometimes even five stretchers in series throughout the length of the courses.

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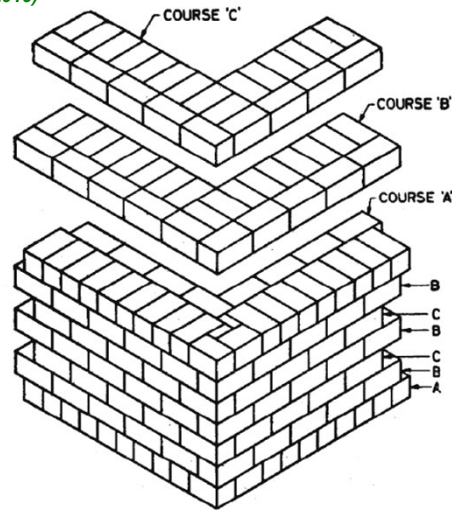
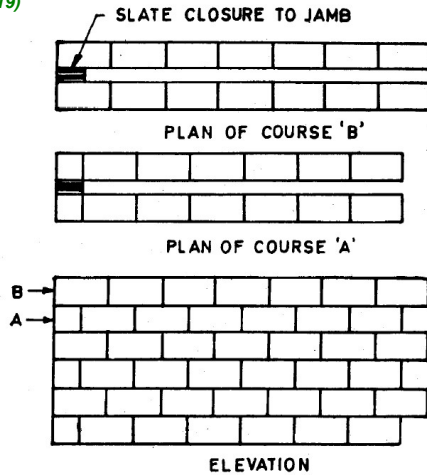


FIG. 7 AMERICAN BOND

- American bond (also known as Common bond) consists of one header course to a number of stretcher courses. The number of stretcher courses may vary from five to seven.

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NOTE — Use of Stretcher bond in cavity wall construction is illustrated here.

FIG. 8 STRETCHER BOND

- In this bond, all the courses are stretcher courses and the overlap is usually half-brick and is obtained by commencing each alternate course with a half-bat.

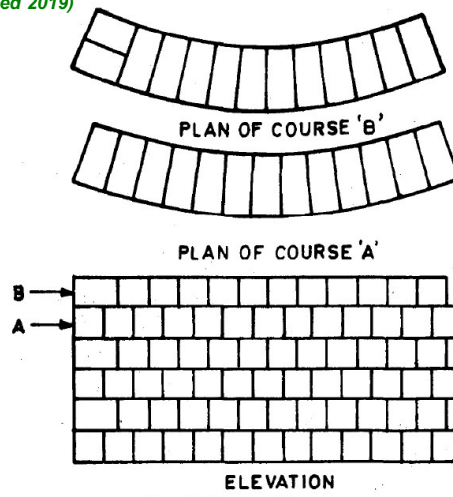


FIG. 9 HEADER BOND

- The facing of this bond has all the courses as headers only and the overlap, which is half the width of the brick is obtained by introducing a three-quarter bat in each alternate course as quoins.

Efflorescence

- ✓ Clay bricks contain soluble salts, which migrate to the surface along with water. As water evaporates to outside air, these salts are collected as patches of white crystals on the face of brickwork. These irregular and unsightly patches are known as efflorescence.
- ✓ The most common salts are calcium sulfate and magnesium sulfate, although various salts of sodium and potassium are also found. These salts originate from the raw clay, and in some cases, are formed during the burning process. The percent of calcium sulfate in bricks is generally very high as compared to other salts.
- ✓ The extent of efflorescence depends on the type and amount of salts. Calcium sulfate is less soluble and takes a longer time to leach out of the bricks. The solubility of magnesium sulfate is greater and may cause flaking of brick faces. Magnesium sulfate is the most destructive soluble salt in the clay units.
- ✓ Efflorescence is more pronounced in exposed walls such as parapets, chimneys, and basement walls.
- ✓ To prevent efflorescence, the walls must be protected from migration of water. Any method to prevent water migration into the walls is helpful in controlling the efflorescence.

References

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